

MECHANICAL-ENGINEERING DEPARTMENT
THEORY EXAMINATION

Month and year of the examination: **May 2025**

Programme: **B.Tech (2nd Sem.) P&I Engg.**

Subject: **Manufacturing Processes**

Course Code: **PIPC-101**

Session: **May/June 2025**

Max. Marks: **50**

Time allowed: **03 Hours**

Note –All questions are compulsory, Attempt the questions in serial order. Write in brief and explain with the help of well-labelled, neat and clean diagrams for scoring good marks.

Q1	(a) Explain the closed mold solidification-based manufacturing process along with the use of core and chaplets with the help of well-labelled, neat and clean diagrams.	5 marks
	(b) Explain the particulate processing method using a diagram only, along with the role of the sintering operation and the sintering temperature.	5 marks
Q2	(i) Differentiate between (a) Boring and Knurling (b) Electrochemical and Thermal Spray coating deposition using diagrams.	5 marks
	(ii) Explain the processes which involve heat treatment and particle impacts to relieve the residual stresses.	5 marks
Q3.	(a) Explain the CVD and Laser Cladding processes with the help of diagrams, and what are their applications.	5 marks
	(b) Explain the drawing and lancing operations performed on sheet metals and what are their applications.	5 marks
Q4.	(a) Give two uses of flux in arc welding, explain the limitations of the shielded metal arc welding (SMAW) process and which process overcomes the limitation. Explain that process with the help of a schematic diagram.	(1+1+3) marks
	(b) (i) Give the complete classification of steel (ii) materials used for cryogenics, excavator buckets, machine tool beds, Pelton wheel buckets, and prosthetic implants.	(2.5+2.5) marks
Q5.	(a) What is the difference between wrought and cast alloys, how will you represent them for Al alloys? (b) Difference between Brass and Bronze, which Cu-alloy offers the highest strength and also writes its application. (c) What is added to lower the fusion temperature and improve the stability of a glass material? Give the composition of CorningWare glass-ceramic. (d) Give two important properties and limitations of polymer material. (e) Give the complete classification of composite materials.	(2×5=10) marks